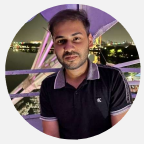
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Priyansh Choudhary

SOFTWARE ENGINEER II | Reports to: Shashank Rachamalla

Connect Nov 2025

Reflection Period: May 7, 2025 - Nov 10, 2025

Impact is **what** you accomplish - and **how** you do it matters just as much.

WHAT

What results have you delivered for your goals, keeping security, quality, and AI in mind?



HOW

How have your behaviors and actions led you, your team and Microsoft to excel, grow, and build trust?

Reflect on the past

What results did you deliver, and how did you do it?

Circle of Impact	What was delivered? How was it delivered?	What Impact/results are accomplished?
Feature	**1. AKS Velero Plugin E2E Tests** • Delivered a comprehensive **E2E validation pipeline** for the **AKS Velero plugin** by	• Led pre-release validation for the 1.17 Azure Blob integration and **resolved a major authentication-breaking issue.**

	designing a **unified release-validation** workflow. • Built an **end-to-end test suite** covering both upstream and Azure-specific scenarios, introduced **daily automated regression** reporting, and fixed multiple **upstream pipeline gaps**. **2. Direct Drive Design & Implementation (In Progress / Post-CCOA Release)** • Designed the **full system architecture (HLD + LLD) for enabling direct-drive backup and restore support across AKS and Disk customers**, covering management, control, and data plane changes. **Aligned Disk partner teams** on VSC flag updates, **resolving a CSI driver bug blocking vaulted restores**, **implemented AAD authentication support**, and performed major refactors across repos. Identified and proposed **fixes for Velero async-backup issues**, and **drove upstream uniformity discussions**. **3. Modify Protection for AKS (Post-CCOA Release)** • Owned the **Modify Protection feature** in response to repeated **customer asks.** Implemented **backend plugin logic and tests**, and **independently built Portal and CLI changes** to accelerate release and reduce cross-team dependencies. **4. Regional Support for AKS Ops and Vault** • Expanded AKS Operational and Vault tiers into 4 new regions by preparing infrastructure, enabling runners, and coordinating with the DM team to close regional gaps for Vault enablement.	• Contributions were acknowledged resulting in **promotion to the Velero Triage role**. • Lays the foundation for reliable DD support that addresses customer needs and **helps toward achieving 99.9 SLA for AKS.** • Allows customers to update protected data sources **without recreating backup items**, **reduces customer friction.** • **Unlocked new customer** deployments, improved geographic availability for AKS Backup.
Quality+ Productivity	**1. Production Release for SLA** Delivered critical production releases (SPH 2505 and Shim 2505) to improve AKS Backup stability and SLA. **Resolved OOM Kill issues** impacting customer workloads, converted multiple **false service errors into accurate user errors**, and **introduced release**	• Raised AKS Operational Tier **SLA to 99.9%.** • **Reduced false alarms** and **improved error accuracy** for operations and support teams.

	notes for Shim to improve coordination with partner teams. **<u>2. SLA Improvement for AKS</u>** Drove AKS hardening efforts through targeted fixes, **manual ReIR for 12 BI**s, and improvements to error classification. Added non-retriable user-error checks, improved **validation for invalid properties**, **fixed restore-from-same-RP** issues, strengthened **Trusted Access calls resilience**, and added **API server error handling**. Identified **systemic customer configuration issues** (CRD misconfigurations) via User Errors. **<u>3. BVTs</u>** Revamped BVT pipelines to **improve resiliency** and operational **reliability**. Fixed **throttling-related failures,** introduced **zonal resiliency**, added **retries for transient** Disk/AKS issues, enabled **dynamic naming** to avoid stale runs, implemented better **cleanup logic**, and **resolved compute quota** blockers. **<u>4. Runners</u>** Stabilized **AKS & Disk, national cloud, and canary runners** by fixing deployment-breaking **CRR changes**, **resolving exception-parsing** issues, **adding resiliency to Configure Protection runners**, fixing long-standing Disk Canary runner issues, and **restoring missing trace logs for better observability.** **<u>5. OSS Contributions</u>** Delivered significant upstream improvements to Velero by **updating authentication libraries**, **implementing newAzureCredential()**, **refining snapshot checkpoint logic**, **adding E2E support for image registry proxies and metadata**, and enhancing **context-based logging**. Maintained **bi-weekly alignment with maintainers** to **represent Azure requirements** and ensure upstream consistency.	• Improved backup and restore reliability, reducing operational noise and support escalations. • Achieved **consistent 99%+ Hardening SLA through** proactive fixes and customer engagement. • **Reduced restore failures stemming from upstream, TA,** and configuration gaps. • Improved **BVT stability**, reducing false failures. • Enabled **consistent test performance across regions** and cloud zones. • **Reduced runner failures, ensuring accurate validation** and early detection of regressions. • Increased Azure's influence in the Velero community by **consistent contributions and collaboration.**
Supportability	**<u>1. Operability</u>** Conducted AKS hardening **bug reviews with Prot, DM team.**	• **Improved accuracy of SLA and fault diagnostics.**

	Fixed false positives in RAS and added **new Vault observability queries**. Fixed **logging gaps in RAS** and **mapped Trusted Access header failures to customer-facing error codes.** **<u>2. CRIs</u>** Fixed **CRR false-positive success** cases, added **explicit errors for unsupported ADLS** accounts, **validation gaps in BMS/BA** corrected. **Supported customer transitions** from SPN to SAMI during Configure Protection failures and **resolved CRR failures** caused by network restrictions.	• **Reduced operational noise** and improved prioritization for engineering teams. • **Simplified troubleshooting** for customers and **helped onboard.** • **Reduced escalations** through **clearer, actionable error messages.**
Security	Resolved **high-risk S360 warnings**, fixed a **credential leak in Geneva entity**, and **sanitized sensitive API keys in Prot AKS Extension logs.**	
Leveraging Others / AI	Used **PR Assistant for small fixes**), applied **MCP Servers to generate design workflows** and diagrams (PV2, Velero Pipeline), used **RTI MCP Agent with Kusto** for automated log analysis and visualizations, **automated weekly SLA/error-code** summaries.	• Significant **reduction in engineering time** on repetitive tasks.
Helping Others	Fixed **Shim release pipeline issues**, assisted with **MDS findings**, **improved Vault stats dashboards**, authored and refined **TSGs for ReIR/CRI/debugging**, **tested Auto ReIR and RA retry flows**, Updated K8s version across all runners, and took SME on-call duty for three continuous weeks.	

Reflect on recent setbacks - what did you learn and how did you grow?

During recent projects, I encountered situations where partner teams or OSS maintainers initially did not agree with our proposed approaches, whether it was design changes for direct-drive backups, Velero async-backup fixes, or other operational improvements. These small setbacks taught me the importance of effectively **convincing others**. I realized that delivering a strong technical solution is not always enough - you also need to **present your ideas in a way that aligns with others' perspectives**, anticipate concerns, and build consensus.

Through these experiences, I learned to approach disagreements strategically: understanding others' priorities, framing my points with supporting data, and identifying win-win compromises. I am committed to further growing this ability to influence, collaborate, and navigate differing viewpoints in complex ecosystems.

Manager comments

Priyansh, you **independently drove** the design and implementation of a **dedicated E2E test pipeline for Velero**, focused on Azure-specific scenarios.
Despite multiple setup challenges, you showed a **strong growth mindset**, learning quickly and iterating to deliver a robust solution without relying on constant guidance.
The pipeline now runs before every Velero release, enabling early detection of issues and providing release sign-offs.
Your work prevented a major regression by catching a critical bug in Velero 1.17 that would have broken Azure Filesystem Backups (FSB).
You also fixed several Azure-specific bugs, significantly improving stability for all users and strengthening Microsoft's OSS presence.
As a result of these impactful contributions, you earned a **Triage role in the Velero repository**, reinforcing trust in your technical leadership within the OSS community.
Outstanding work! Continue contributing actively to OSS, and aim to grow into a **full contributor role in the Velero repo**.

Some feedback from your peers: "You have done a fantastic job with the OSS upstream E2E test pipelines. Your approach to fix issues upstream and identify P0 Azure bugs demonstrates a true growth mindset"

Priyansh, you **independently drove the PV2 design and implementation** for AKS backup, owning the full system architecture (**HLD + LLD**) and aligning partners across CSI and Velero. You resolved a critical CSI driver blocker by relaxing snapshot size checks to support PV2 restores and proactively implemented **AAD authentication support** for Disk snapshots during AKS tiering, fixing audience URL issues and enabling the flow for customers.
Your **data-driven approach** brought clarity to the PV2 strategy, simplifying technical implementation while ensuring an optimal end-user experience.
With PV2 introducing Velero's **async handler for the first time**, you identified gaps in async workflows related to updating backup metadata and successfully closed them through upstream community discussions.
Great work ! Really looking forward to bringing PV2 support for AKS to prod.

Some feedback from your peers: I want to highlight your excellent work in driving clarity for the PV2 approach for AKS backup. You collaborated effectively with partners across CSI and Velero !

Priyansh, you **proactively picked up the Modify Protection feature** as part of the hackathon, addressing a top customer ask with urgency and ownership. You implemented the **backend plugin logic and tests**, and went beyond boundaries by independently building **Portal and CLI changes**, accelerating delivery and reducing cross-team dependencies.
Your energy and commitment were evident as you completed and demoed the feature during the hackathon, showcasing end-to-end execution.
This work reflects your ability to take initiative, work across layers, and deliver impactful outcomes under tight timelines.
Great job! Looking forward to thorough testing and shipping this to production soon.

Priyansh, you have demonstrated exceptional diligence in driving **operability excellence**. You consistently stayed on top of critical issues, especially in the **AKS vault tier**, ensuring timely resolution and minimal customer impact. Your proactive

collaboration with multiple internal stakeholders and strategic escalations built the right traction to unblock progress. Through your consistent efforts, we maintained SLA commitments and improved overall product reliability. This level of ownership and responsiveness sets a high bar for operational rigor.
Keep up the great work and continue being a role model for operability within the team.

Great to hear about your learning, and I completely agree. It's crucial to present your ideas with clarity, back your questions with strong logic, and keep conversations crisp and focused."

Plan for the future

What are your goals for the upcoming period?

Goal #1. Direct Drive Support Release

- Aim to **GA the Direct Drive backup and restore support** for AKS and Disk customers as soon as possible.
- Target **elevating SLA to 99.9%+** for all affected workloads.
- Ensure **zero customer-facing regressions** by thoroughly testing and validating all failure points before release.
- Proactively identify and **fix any potential failure scenarios**, including edge cases discovered during BVT, staging, or early adopter testing.

Goal #2. Velero 1.15 Merge

- Pull **Velero upstream 1.15** and merge it into Azure's pipelines.
- **Integrate existing E2E and CI/CD pipelines** to maintain reliability while adopting new upstream changes.
- Improve **security governance, logging, and performance** through features and updates introduced in the 1.15 release.
- Maintain a **consistent and smooth experience for customers** after the upgrade, with zero functional regressions.
- Ensure **CG-compliant downstream changes** for Azure-specific extensions and integrations.

Goal #3. Expanding Testing Frameworks for Feature Reliability

- Design and implement **scale test scenarios for vault tier** including restoring very old backups, recent backups, backups with churned data disks, and scenarios involving up to 100 disks per backup, to validate resilience at scale.

- Enhance existing BVT coverage by introducing variations that reflect real-world customer workloads and edge cases.
- Continue collaboration with stakeholders to align test efforts with evolving feature goals and usage patterns.
- Achieve **100% SLA adherence** for daily BVT execution and actionable issue reporting.
- Ensure **>90% coverage of prioritized and GA-bound features in BVTs** by the end of the quarter.

Goal #4. Contribute to and Develop from Open-Source Initiatives

- Actively **pitch design-oriented solutions** to Velero, to address operational gaps in Object-Locking.
- Contribute important feature to **Velero 1.19 Release.**
- Work towards achieving **active contributor status**, establishing credibility and recognition in the Velero community.
- Align **Azure-specific requirements** with upstream development to ensure maintainability and long-term compatibility.

How will your actions and behaviors help you reach your goals?

- **Proactive Ownership:** I will take full ownership of the Release, ensuring that all dependencies are identified and managed early.
- **Early Detection of Risks:** Conduct thorough testing in staging and BVT environments, Bug Bashes to identify potential failure points before GA.
- **Collaboration:** Coordinate with Disk, AKS, and Portal teams to ensure smooth integration and resolve blockers quickly.
- **Continuous Monitoring:** Implement and analyze metrics and logs to ensure SLA targets are met, and regressions are prevented.

Upstream Consistent Engagement: Regularly sync with maintainers, participate in discussions, and provide thoughtful PRs to build credibility.

Manager comments

Looks good Priyansh. Continue to drive the impact by completing PV2, Modify protection work without any regressions.